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# Conversational Recommender System for Vehicles

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**Abstract**

In this report, we design a Conversational Recommender System for vehicle selection. The system helps users express their preferences in natural language and translates them structured technical terms. We propose a dual data strategy that combines two data retrieval approaches. The first is constraint-based filtering over a structured SQL-like database of vehicle attributes. The second is FAISS-based similarity search over vector embeddings generated from vehicle brochures. We evaluate the proposed system against multiple variants. LLM such as Qwen and LLama without any data retrieval represents the baseline. More advanced comparison represent a non-conversational version of the proposed recommender system. In all cases the proposed recommender system outperformed the others.

**TODO! Dodaj dejanske rezultate al neki**

**Keywords**

Conversational Recommender System, RAG, FAISS, Car recommendation, LLM

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## Introduction

In the past years, choosing a suitable vehicle has become very challenging due to a wide range of options. Customers can choose between many models with different fuel types, engine characteristics, size, and other specifics.

Most existing systems that assist users in selecting vehicles are based on predefined filters, such as price range, brand, transmission types, and others. While this approach is effective for experienced users, it might not be that suitable for those who cannot easily translate their needs into technical details.

As an improvement to such systems, Conversational Recommender Systems (CRS) have been introduced. A CRS is an interactive system that allows users to express their preferences through dialogue, while the system can ask follow-up questions to refine these preferences and provide the best recommendation(s).

In this paper, we propose a hybrid CRS for vehicle selection using that combines large language models (LLMs), structured constraint extraction, and retrieval-augmented generation (RAG). The system engages the user in a conversation with an LLM that helps them express and refine their vehicle preferences. The final product is a system that simplifies making a decision for non-expert users, and also improves customer experience in vehicle selling companies.

## Related work

Several studies have explored recommender systems, however the vehicle domain remains limited. We categorized existing work into recommendation systems for vehicles and conversational recommender systems for various products.

In the context of vehicle recommendation, Beddouda et al. [1] proposed a system using BERT4Rec, which predicts which cars a user might be interested in based on past interactions (without any user communication). Singh et al. [2] introduced a one-turn dialogue system that uses LDA model to analyse user reviews, where users describe their car preferences and the system returns top few recommendations.

For general CRS, several multi-turn dialogue systems have been proposed, especially in the travel domain. Liao et al. [3] proposed a deep recommender system using seq2seq with a neural latent topic component. Baizal et al. [4] introduced an ontology-based system, and Carrillo et al. [5] introduced a system that uses Chatbot to collect user preferences and generates similarity-based recommendations using SBERT embeddings.

In addition to the travel domain, CRS have been implemented in several other domains. Sun et al. [6] developed a reinforcement learning based CRS for restaurant recommendation using Yelp review data. Zhu et al. [7] proposed a movie recommendation CRS using retrieval-augmented generation (RAG) approach in combination with a large language model.

## Methods

The proposed system is designed as a Conversational Recommender System (CRS) that integrates large language models (LLMs), structured data processing, and retrieval-augmented generation (RAG) to provide personalized vehicle recommendations through natural language interaction. The overall pipeline is described below as well as shown on the figure 1.

### Data Preprocessing

The first dataset was extracted from the CarAPI website [8]. It provides vehicle data in JSON format, such as price, fuel consumption, engine specifications... Data was then processed and stored into structured SQLite database (on Figure 1 marked as Structured database).

The second dataset consists of vehicle brochures in PDF format extracted from the Auto-Brochures website [9]. The dataset includes multiple car brands available on the US market, of which 26 were selected for this work. We used a Qwen embedding model to generate vector represenations of the dataset content. These embeddings were then indexed using FAISS (Facebook AI similarity search) [10] to enable efficient similarity search (Dataset is marked on Figure 1 as FAISS Vector Database).

### Conversational Recommender System - CRS

The system begins with user describing in natural language their preferences and requirements they look for in a new car. That input is then parsed by the LLM (on figure 1 marked as LLM Parser).

Parsing is implemented by first extracting relevant attributes from the user's query using the embeddings of the description of the attributes. When we defined a schema for the structured database, we also crafted metadata for each attribute in a form of a short description. This description also includes synonyms, related terms, user intentions and example queries. Then we embed those descriptions using an embedding model, which was selected with benchmarking of different models, with a specialised instructions for the task. When the user query is received, we embed it using the same embedding model and then calculate similarity between the query embedding and the attribute description embeddings. The attributes with the highest similarity are selected as relevant to the user's query.

For each relevant attribute, the LLM is then prompted to extract the corresponding value and constraint, which proved to be more effective than directly prompting the LLM to extract every attribute at once.

Table 1 shows an example of the extracted structured information from a user query. The LLM is able to identify that the user is looking for an SUV with electric engine, but it cannot determine the exact value of the fuel consumption constraint, which is then marked as None.

| Name | Value | Constraint (min, max, equal...) |
|---|---|---|
| body_type | SUV | equal |
| engine_type | electric | equal |
| combined_l_per_100km | None | None |

**Table 1.** Example of parsed text for the following user's query: "I'm looking for a spacious SUV with low consumption, preferably electric."

That is why a single user query is usually not enough to generate a good recommendation. User also might not express all their preferences in a single query, leading to a very large choice of cars.

To address these issues, we improved the system with creating a conversational LLM. (marked on Figure 1 with the same name). This model interacts with the user to complete missing constraints and suggest additional attributes that could be in interest to the user.

Then the system employs a dual data strategy using databases described in the previous section. First, we fetch relevant cars from the database using constraint-based filtering, which yields the first candidate set. Next we embed the user's query, along with the system's conversation history, and use it for semantic search in the FAISS vector database. Which contains embedded chunks from the vehicle brochures. This allows the system to retrieve cars that are semantically similar to the user's preferences, even if they do not exactly match the structured constraints. While we can extract information such as number of seats, or fuel type from the structured database, other information such as a car's feel, design or equipment is not easily captured in structured form, but can be retrieved from the brochures using the semantic search.

The system then combines both retrieved sets of car candidates and filters them again to ensure that candidates from the FAISS search also meet the user's constraints. The contex of the retrieved chunks is then correctly formatted and provided to the LLM as part of the final recommendation prompt. So the LLM can summarize the retrieved information and generate a final recommendation response. The LLM also comments on the key differences between the suggested cars, which can help users make an informed decision.

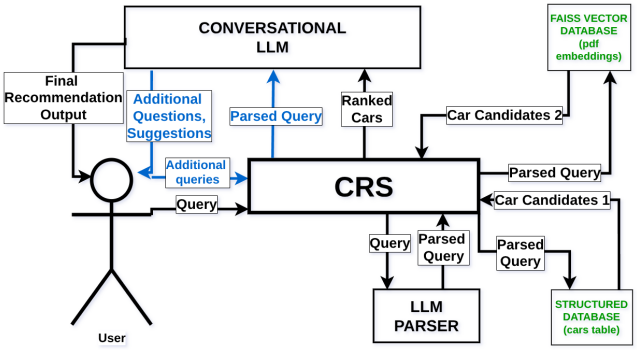


**Figure 1.** CRS Architecture. Blue arrows represent conversation interaction loop between the user and the LLM, green components represent databases that the system uses.

### Evaluation methods

**Attribute embeddings evaluation.** We designed a benchamrk to evaluate the performance of different embedding models for the task of identifying relevant attributes from user queries. We created a dataset of 40 user queries, each manually annotated with the relevant attributes. We then tested several embedding models, from Qwen, BAAI-bge, nomic-ai, GTE, Google's gemma and jinaai. To rank the models we measured Spearman's correlation to measure how well the model's ranking of attributes agrees with the ground truth. We also measured the separation with Cohen's $d$, since we want a large distance between the relevant and irrelevant attributes, and finally we also measured the F1 score, which was also optimized with different thresholds. Results of the benchmark are ranked embedding models with their optimal thresholds. The best model was `Qwen3-Embedding-4B`, with `jina-embeddings-v5-text-small` in the close second place. We however used **`Qwen3-Embedding-0.6B`** which ranked 3rd but had the best performance compared to its speed.

**TODO: add LLM's names**

Evaluating language models is challenging because their outputs are not fully deterministic, especially when they are in conversation mode. We designed three different tests to evaluate the system and the models.

First, we evaluate whether the system performs better than a baseline. We manually created a set of user queries and manually parsed them into constraints (similar to Table 1). We then ran our system in non-conversational mode, which directly returns a set of recommended cars. This result is compared to the set of cars obtained from the structured database using the manually verified constraints. From this comparison, we compute standard metrics such as number of correctly retrieved cars, missed cars, precision, recall, and F1 score. We also run the same evaluation for a baseline LLM without any system support.

The second evaluation tests the conversational setup. This evaluation is not fully automated. We use the same queries as before, but simulate multi-turn interaction by providing follow-up responses after each LLM question. After the conversation ends, we store the final set of recommended cars and compare it to the expected set obtained from the structured database using the manually parsed query. We also join user's multiple queries into one and send them to the non-conversational and baseline LLM. It is not fully fair, but we also compare evaluation of our main pipeline with these two.

The third evaluation tests whether the RAG component is working correctly. After each recommendation, we manually check whether the returned cars satisfy the user's stated preferences.

## Results

## Discussion

Use the Discussion section to objectively evaluate your work, do not just put praise on everything you did, be critical and exposes flaws and weaknesses of your solution. You can also explain what you would do differently if you would be able to start again and what upgrades could be done on the project in the future.

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